Extended reading — Week 03: Monte Carlo Methods

October 6, 2026

How to use this list

The first paper is the source of every stochastic-approximation guarantee in RL. The second is the resolution of the first-visit vs every-visit MC question. The third is the canonical reference for per-decision IS (the variance-reduction trick that makes off-policy RL tractable). The fourth is the modern reference for off-policy evaluation with high-confidence bounds, which you should read *before* deploying any new policy. The fifth is the bridge from off-policy MC to deep learning.

1 Robbins & Monro (1951) — A stochastic approximation method

Ann. Math. Stat. 22, 400–407.

Summary. Introduces the stochastic-approximation iteration $\theta_{n+1} = \theta_n + \alpha_n(Y_n - f(\theta_n))$ for finding the root of an unknown regression function. Proves convergence in mean-square if $\sum \alpha_n = \infty$ and $\sum \alpha_n^2 < \infty$ — the celebrated "RM conditions".

Why it matters for this week. The incremental sample-average $Q_{n+1} = Q_n + \frac{1}{n}(R_n - Q_n)$ is exactly the Robbins–Monro algorithm with $\alpha_n = 1/n$. The convergence proofs of every "learning rate goes to zero" RL algorithm (SARSA, Q-learning, TD(λ) on-policy) hinge on Robbins–Monro.

> *"Convergence in mean-square is established under conditions that include $\sum \alpha_n = \infty$, ensuring the iteration moves forever, and $\sum \alpha_n^2 < \infty$, ensuring the noise averages out."*

"Move forever, but slow down quickly enough that noise averages out" is the single most important piece of intuition for learning-rate selection.

2 Singh & Sutton (1996) — Reinforcement learning with replacing eligibility traces

Machine Learning 22, 123–158.

Summary. Compares first-visit and every-visit MC and shows the bias of every-visit is $O(1/N)$. Introduces *replacing* eligibility traces (overwrite, not accumulate) and proves they correspond to first-visit MC in the $\lambda = 1$ limit. Sets up the unification of MC and TD via λ that Week 4 builds on.

Why it matters for this week. It is the reason "first-visit" shows up as the default in MC reference implementations. Read §3 of the paper for the cleanest treatment of the bias issue. The same paper also explains the otherwise mysterious choice "replacing" vs "accumulating" traces in classical TD(λ).

> *"First-visit MC produces independent samples of the return per state per episode; every-visit MC produces dependent samples but does not waste data. The bias of every-visit vanishes as* $1/N$ *in the number of episodes."*

The "does not waste data" is the practical case for every-visit; the "independent samples" is the theoretical case for first-visit. In most deep-RL implementations the distinction is academic — the variance is dominated by other factors.

3 Precup, Sutton & Singh (2000) — Eligibility traces for off-policy policy evaluation

ICML 2000.

Summary. Defines per-decision importance sampling for MC and TD methods. Proves that the per-decision estimator has the same expectation as the trajectory-level OIS but lower variance whenever the immediate rewards depend on the action at time t but not on later actions. Generalises to TD(λ) with off-policy targets.

Why it matters for this week. If you remember one off-policy trick, remember per-decision IS. The trajectory-level estimator is unbiased but exponentially noisy in the horizon T; per-decision IS keeps the unbiasedness but only multiplies by

the importance ratio *up to* time t for each reward R_{t+1}. The variance reduction is dramatic in any long-horizon setting.

> *"Per-decision importance sampling reduces variance without introducing bias by noticing that the reward at time t is independent of the actions taken after time t given the policy."*

That single observation underpins every modern off-policy estimator. It is also what makes truncated trust-region methods possible (PPO, GRPO).

4 Thomas & Brunskill (2016) — Data-efficient off-policy policy evaluation for reinforcement learning

ICML 2016.

Summary. Introduces the *doubly-robust* estimator for RL: combine importance sampling with a learned reward / value model so the estimate is consistent if *either* the model or the propensity is correct. Provides finite-sample concentration bounds and high-confidence off-policy evaluation: a $(1-\delta)$ lower bound on V^π that can be inspected before deployment.

Why it matters for this week. Read this before deploying any candidate policy to production. The DR estimator is the right default in any applied setting with both logs and a reasonable reward model. The high-confidence lower bound (Section 4.3) is the deployment gate: only ship a policy whose lower bound exceeds the production policy's upper bound.

> *"A safe policy improvement scheme: never deploy a policy whose 95% lower bound on value is below the current policy's empirical value. Doubly-robust estimation provides the bound that makes this protocol tight."*

"Never deploy unless the lower bound dominates" is the cleanest production rule in RL. It is also the rule that, if synthetic insurance / ATS had it on day one, would have prevented several historical regressions.

5 Joachims, Swaminathan & de Rijke (2018) — Deep learning with logged bandit feedback

ICLR 2018.

Summary. Trains a deep neural-net policy on logged bandit feedback using the SNIPS estimator as a counterfactual risk objective. Adds *propensity overfitting* controls (self-normalisation, weight clipping) to keep deep counterfactual training stable. Bridges classical off-policy MC to modern offline-RL methods like CQL.

Why it matters for this week. It is the deep-learning analogue of your week-3 workbook exercise: estimate the value of a target policy from logs and *optimise* the policy against that estimate. The propensity- overfitting analysis is exactly the safety concern of the lecture.

> *"Self-normalisation defeats propensity overfitting — the failure mode where the policy learns to put mass on rare-but-high-weight logged examples. We demonstrate this failure empirically and show that SNIPS-based training avoids it."*

"Propensity overfitting" is the deep-learning name for the OIS variance explosion. Knowing both names lets you talk to both communities.

Reading exercises

1. **Robbins–Monro, optimal α_n.** Robbins–Monro prove convergence under any α_n satisfying the two summability conditions. The optimal rate $\alpha_n = c/n$ achieves $1/\sqrt{n}$ mean-square error. Read Polyak & Juditsky (1992, *SIAM J. Control*) on iterate averaging — show that averaging the iterates of a slower-than-optimal step size recovers $1/\sqrt{n}$ asymptotic variance.

2. **Singh–Sutton, finite-sample comparison.** Run first-visit and every-visit MC on a chain MDP with $N \in \{10, 100, 1000\}$ episodes. Plot the bias and variance of each as a function of N. Verify the $1/N$ bias claim for every-visit.

3. **Precup–Sutton–Singh, per-decision IS on chains.** Implement per-decision IS on a 5-state chain. Compare its variance with trajectory-level OIS at $T = 20$. Verify that the variance ratio scales like $T \cdot \mathbb{E}[\rho^2]$ vs $\mathbb{E}[\rho_\tau^2]$.

4. **Thomas–Brunskill, doubly-robust on Mountain Car.** Train a small reward model on logs of a sub-optimal policy and use it in the DR estimator. Compare the high-confidence lower bound of DR with WIS on a Mountain-Car-like environment.

5. **Joachims et al., BanditNet on synthetic clicks.** Reproduce Algorithm 1 from the paper on the synthetic click dataset in their Section 5. Compare with a naive cross-entropy classifier; how much does propensity overfitting cost the naive approach?

Connections to later weeks

- *Week 4 (TD)* replaces the MC return by a bootstrap target; IS still applies but per-decision.
- *Week 11 (contextual bandits + OPE)* unifies SNIPS, DR, SWITCH, and MIPS in a single contextual-bandit benchmark.
- *Week 14 (offline RL)* extends doubly-robust to full RL (Jiang & Li 2016) and is the foundation for conservative offline Q-learning.
- *Week 19 (DPO and friends)* uses an importance-weighted contrastive loss directly on preferences — a direct descendant of the off-policy MC ideas of this week.

Monte Carlo is conceptually simple but its lessons (coverage, importance weights, doubly-robust safety) recur in every off-policy method that follows.

Sourced from

`research/bandits_matching_sota.md` §3 and Sutton & Barto Chapter 5. For a deeper dive into OPE see Voloshin, Le, Jiang, Yue *Empirical Study of Off-Policy Policy Evaluation for RL* (NeurIPS 2021 benchmarks).